

Lab 2: Keys and Normalization

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1 Objectives

The purpose of these exercises is to go through the normalization forms within database design. A given problem represents a database design that needs to be normalized. Your objective is to bring it to a higher NF while motivating your decisions.

2 Reading

To be able to actively participate in this lab the student is expected to have read the lectures. Additional reading from provided resources is beneficial but not crucial to the completion of this lab.

3 Setup

There are no special requirements when it comes to tools used. You can either do it on paper or digitally. The solution should be uploaded in a digital format, scan or another type.

4 Presentation

The presentation for Lab 2 is done orally. **All group members** present to the lab assistant which then determines if the demonstrated knowledge is sufficient for a pass. If not, the students can present at a later occasion, same or other day. The lab is graded individually, meaning that one team member can pass while the other fails.

5 Instructions

You must work in a **group of 2 students**, together, but with equal work distribution. Your team will be the same for all the remaining laboratory assignments as well. You should be with a partner that is in the same lab session as you (easier for everybody).

Based on the scenarios provided you will follow the normalization rules and bring the database to 3NF.

We expect you to:

- Bring a problem to 3NF
- Manage keys during the process
- Elaborate what ambiguities there are
- Identify anomalies, remove them, elaborate
- Motivate your decisions

6 Scenario 1

Given the relation R_0 and the functional dependencies F_0 :

- a) Identify what anomalies might occur in R_0 , elaborate
- b) Perform a normalization until 3NF.

$R_0 = \{\text{invoiceNo}, \text{invoiceDate}, \text{customerNo}, \text{customerName}, \text{itemName}, \text{itemPrice}, \text{qty}\}$

$F_0 = \{\text{invoiceNo} \rightarrow \{\text{invoiceDate}, \text{customerNo}\}, \text{customerNo} \rightarrow \text{customerName}, \text{itemName} \rightarrow \text{itemPrice}, \{\text{invoiceNo}, \text{itemName}\} \rightarrow \text{qty}\}$

The candidate key is a combination of $\{\text{invoiceNo}, \text{itemName}\}$

7 Scenario 2

We want to create a "simplified" database for a bank:

You are responsible for designing a relational database for a banking system that manages customers (first name, last name, email, phone, and birthdate), their accounts (current amount, and creation date), and associated transactions (type: "deposit" or "withdrawal", and amount). Draw the Entity-Relationship Diagram (ERD) knowing that each Customer can own zero or more Accounts, an account is for one Customer only. Each Account can be linked to zero or more Transactions and a transaction is only for one account. Your task is to translate this ERD into a well-structured logical model (tables).

Once you have your tables, identify the functional dependencies and normalize the database to 3NF.

8 Scenario 3

Recall: the closure of an attribute (or set of attributes) " u " in a relation R_0 is the set of all the attributes you can reach starting from " u " (including itself) and following the dependencies in F_0 .

We have the relation $R_0 = \{A, B, C, D, E, F, G\}$

and the functional dependencies $F_0 = \{ \{B\} \rightarrow \{A, F\}, \{C\} \rightarrow \{G\}, \{F\} \rightarrow \{E, D\}, \{G\} \rightarrow \{B\} \}$

1. What is the closure of C , $\{C\}^+$?

$\{C\}^+ =$

2. What can be chosen as the primary key for this relation? Motivate your choice.